Andreas Köpf, Yannic Kilcher, Dimitri von Rütte, Sotiris Anagnostidis, Zhi-Rui Tam, Keith Stevens, Abdullah Barhoum, Nguyen Minh Duc, Oliver Stanley, Richárd Nagyfi, Shahul ES, Sameer Suri, David Glushkov, Arnav Dantuluri, Andrew Maguire, Christoph Schuhmann, Huu Nguyen, and Alexander Mattick. 2023. OpenAssistant Conversations – Democratizing Large Language Model Alignment. ArXiv:2304.07327 [cs].

Brenden M. Lake and Gregory L. Murphy. 2020. Word meaning in minds and machines. *Psychological review*.

Clara Meister, Gian Wiher, Tiago Pimentel, and Ryan Cotterell. 2022. On the probability–quality paradox in language generation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 36–45, Dublin, Ireland. Association for Computational Linguistics.

Tomas Mikolov, Ilya Sutskever, Kai Chen, Gregory S. Corrado, and Jeffrey Dean. 2013. Distributed representations of words and phrases and their compositionality. In *Neural Information Processing Systems*.

George A. Miller. 1995. WordNet: a lexical database for English. *Communications of the ACM*, 38(11):39–41.

Shima Rahimi Moghaddam and Christopher J. Honey. 2023. Boosting Theory-of-Mind Performance in Large Language Models via Prompting.

Jay A. Olson, Johnny Nahas, Denis Chmoulevitch, Simon J. Cropper, and Margaret E. Webb. 2021. Naming unrelated words predicts creativity. *Proceedings of the National Academy of Sciences*, 118(25).

OpenAI. 2022. Introducing ChatGPT.

OpenAI. 2023. GPT-4 Technical Report. ArXiv:2303.08774 [cs].

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744.

Yang Jeong Park, Daniel Kaplan, Zhichu Ren, Chia-Wei Hsu, Changhao Li, Haowei Xu, Sipei Li, and Ju Li. 2023. Can chatgpt be used to generate scientific hypotheses?

Jeffrey Pennington, Richard Socher, and Christopher Manning. 2014. Glove: Global Vectors for Word Representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1532–1543, Doha, Qatar. Association for Computational Linguistics.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9.

Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. 2022. Hierarchical text-conditional image generation with clip latents.

Mark A. Runco and Garrett J. Jaeger. 2012. The Standard Definition of Creativity. *Creativity Research Journal*, 24(1):92–96.

Tobias Schnabel, Igor Labutov, David Mimno, and Thorsten Joachims. 2015. Evaluation methods for unsupervised word embeddings. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 298–307, Lisbon, Portugal. Association for Computational Linguistics.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. 2023. The curse of recursion: Training on generated data makes models forget.

David Silver, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, Sander Dieleman, Dominik Grewe, John Nham, Nal Kalchbrenner, Ilya Sutskever, Timothy Lillicrap, Madeleine Leach, Koray Kavukcuoglu, Thore Graepel, and Demis Hassabis. 2016. Mastering the game of go with deep neural networks and tree search. *Nature*, 529(7587):484–489. Number: 7587 Publisher: Nature Publishing Group.

Douglas Summers-Stay, Clare R. Voss, and Stephanie M. Lukin. 2023. Brainstorm, then Select: A Generative Language Model Improves Its Creativity Score. In *The AAAI-23 Workshop on Creative AI Across Modalities*.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. LLaMA: Open and Efficient Foundation Language Models. ArXiv:2302.13971 [cs].

Hugh Zhang, Daniel Duckworth, Daphne Ippolito, and Arvind Neelakantan. 2021. Trading Off Diversity and Quality in Natural Language Generation. In *Proceedings of the Workshop on Human Evaluation of NLP Systems (HumEval)*, pages 25–33, Online. Association for Computational Linguistics.

Jingyi Zhang, Kaixiang Zhuang, Jiangzhou Sun, Cheng Liu, Li Fan, Xueyang Wang, Jing Gu, and Jiang Qiu. 2023. Retrieval flexibility links to creativity: evidence from computational linguistic measure. *Cerebral Cortex*, 33(8):4964–4976.

A Selecting sample size for each model

Figure 7 shows the sample size to generate stable results using top-p sampling for each model. With confidence coefficient $\alpha = 0.05$, standard deviation $\hat{\sigma}$ and error $\epsilon = 1$, we choose $N > (\lambda_\alpha \times \hat{\sigma}/\epsilon)^2 = 1.96^2 \times \hat{\sigma}^2$. We find larger models (GPT-4 and GPT-3.5-Turbo) generate answers more stably.

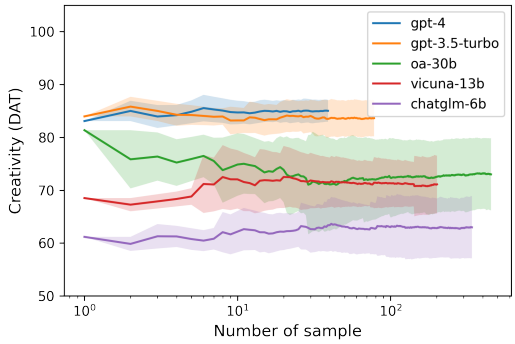

Figure 7: Results of the DAT across different sample sizes. The bands indicate the standard deviations.

B Results of the DAT and surprisal on human and random and baselines

Figure 8 shows the results of the DAT and surprisal on human and random baselines. We find positive relationship between surprisal and the DAT. Random baseline is a strong baseline that has maximal remote association (uniform distribution) despite without inhibition. Even so, we find some people surpass random baseline and approach ceiling DAT at specific section of surprisal.

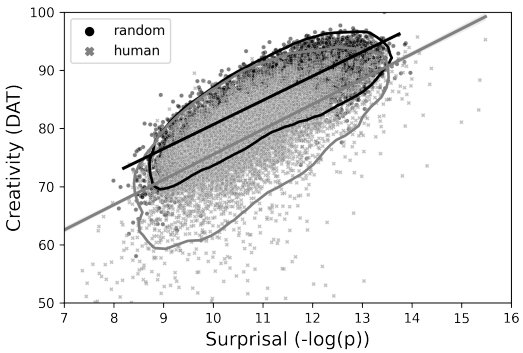

Figure 8: Results of the DAT and surprisal on human and random baseline. Contours indicate the 95% confidence intervals.

C Result of the DAT and surprisal for more models using greedy search

Figure 9 shows the results of the DAT and surprisal for more LLMs using greedy search.

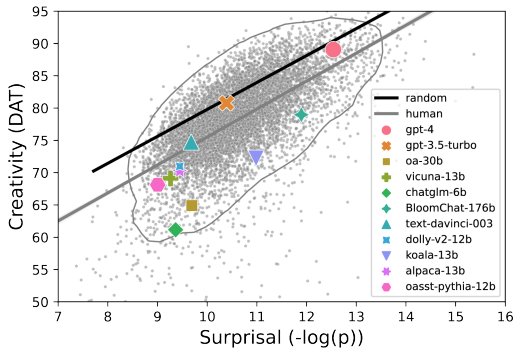

Figure 9: Results of the DAT and surprisal for more LLMs. The contour indicate the 95% confidence interval.

D Controlling surprisal as a confounding variable

Considering the potential influence of word frequency on measuring semantic distance, we control surprisal (Figure 10). The results are similar as before that GPT-4 and GPT-3.5-Turbo outperform average human performance, while other models are below average human level. However, GPT-4 as well as Oasst-Llama-30B lose their superiorities because their high DAT scores partially depend on generating rarer words.

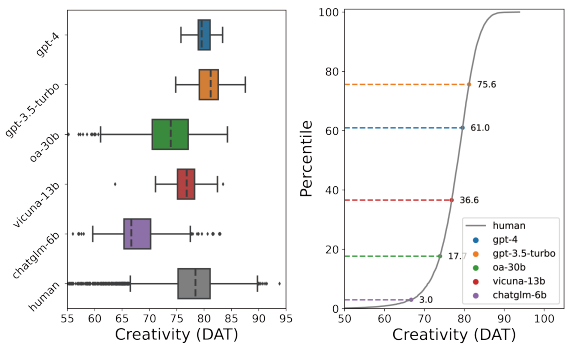

Figure 10: Results of the DAT when controlling surprisal as a confounding variable.